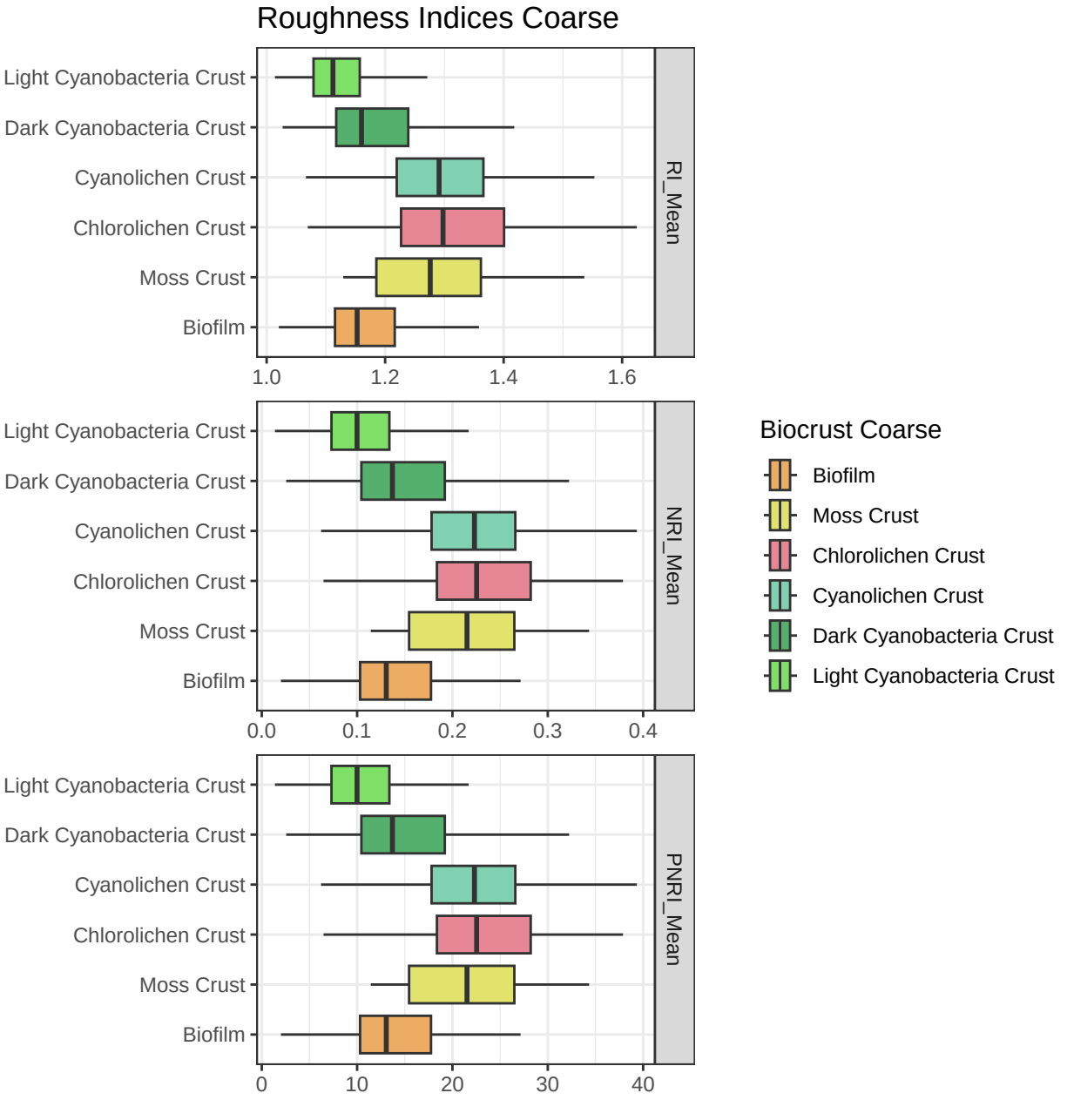
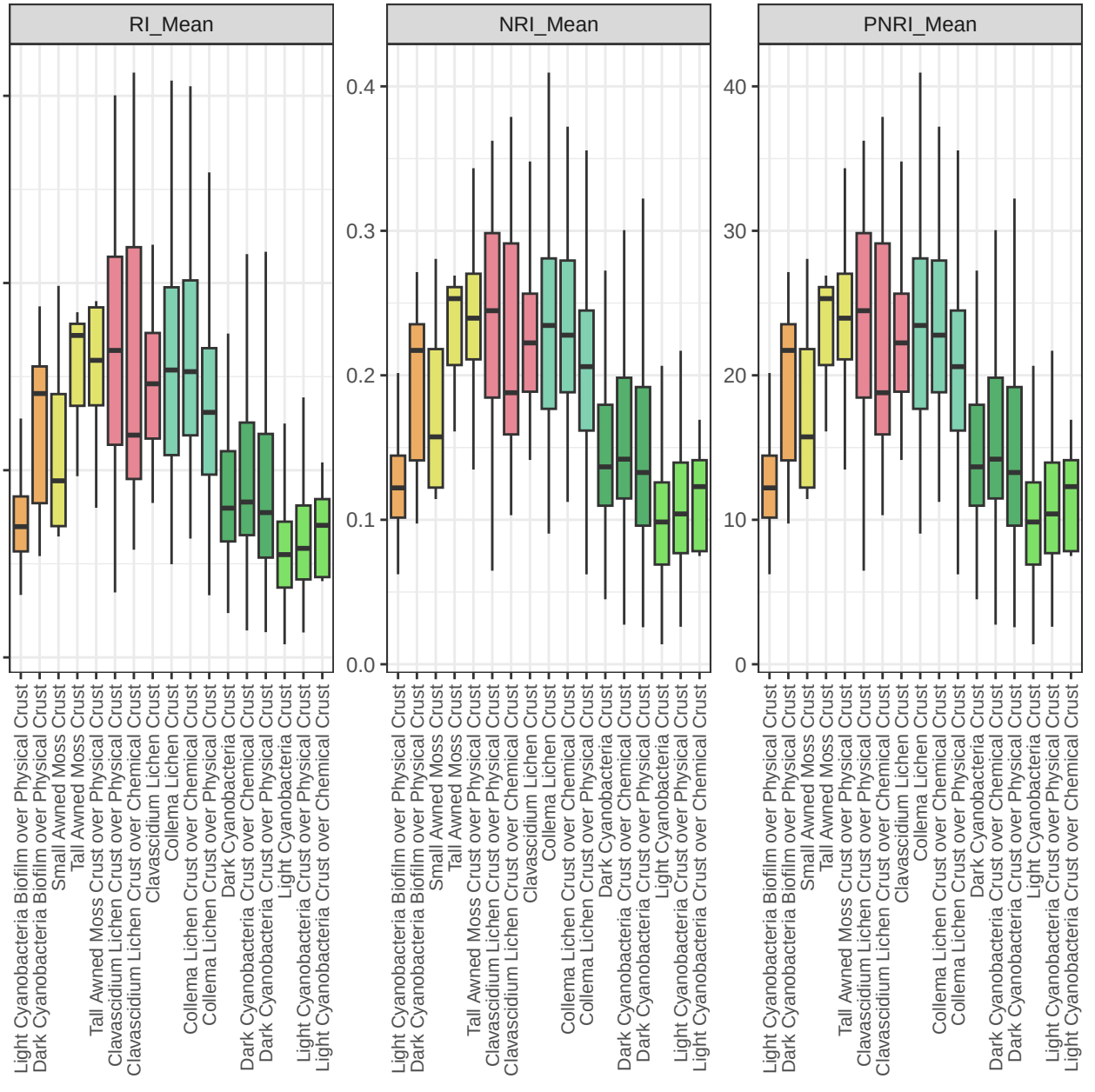
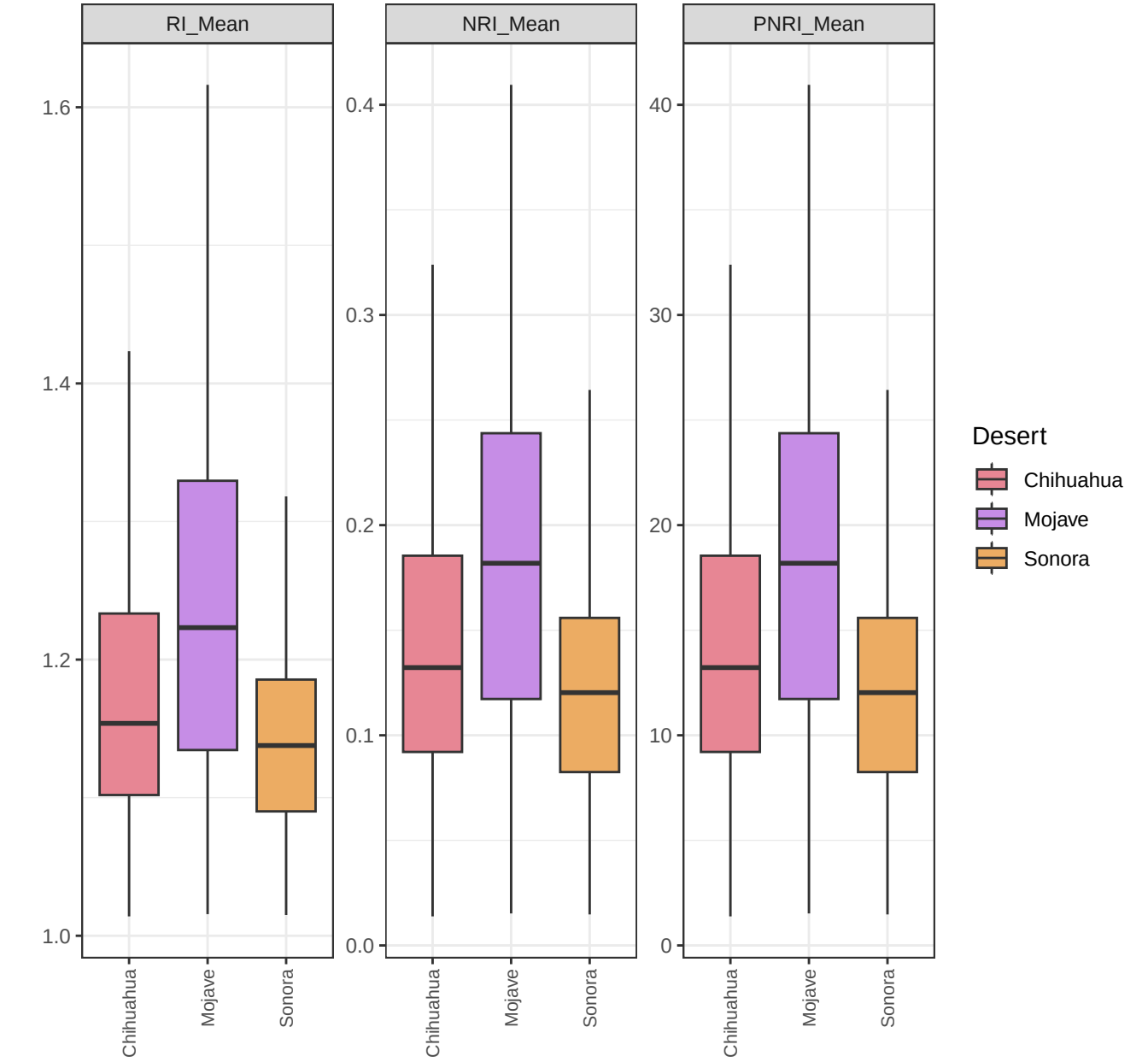
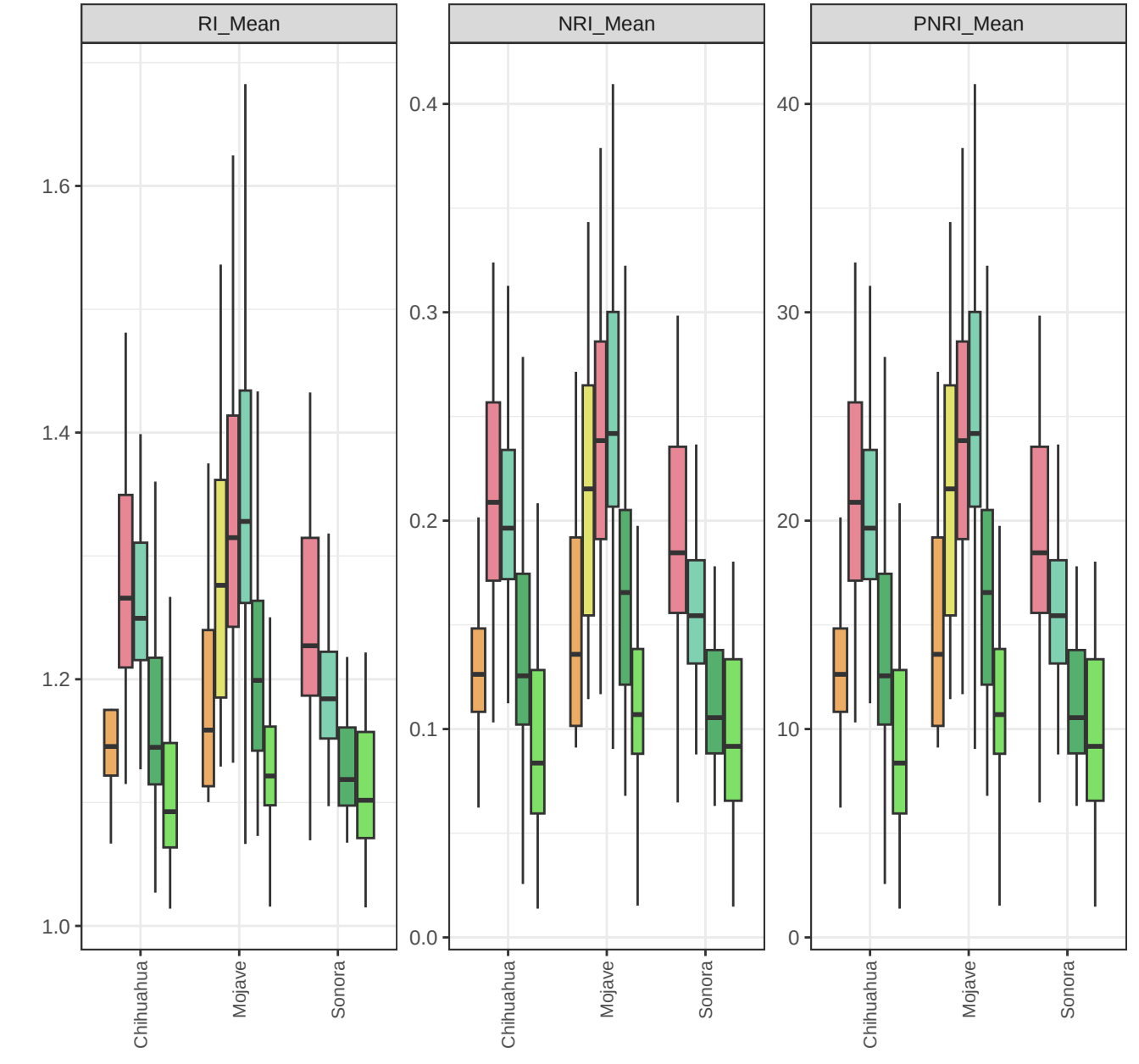
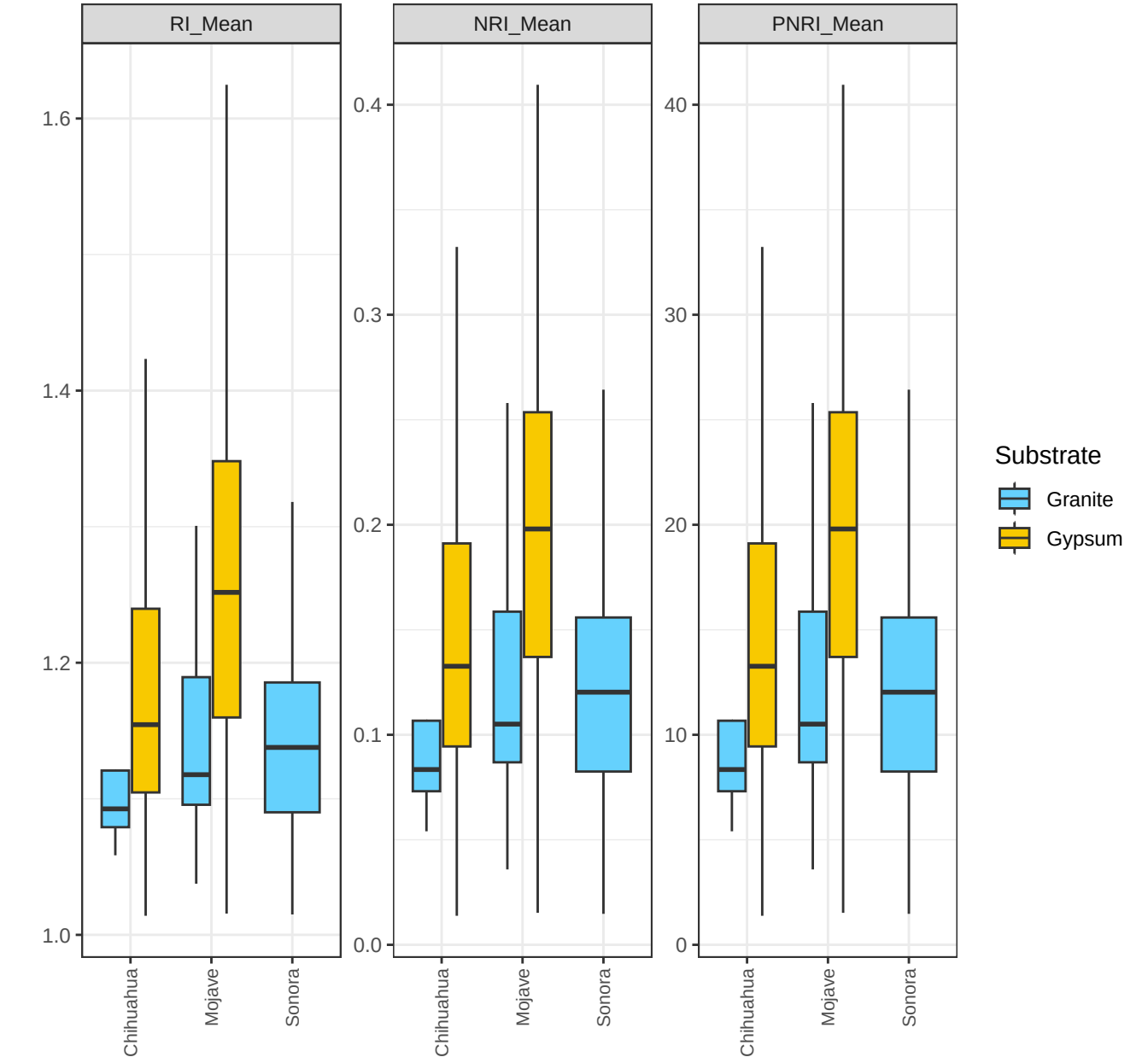

Roughness Indices Coarse
Light Cyanobacteria Crust
Dark Cyanobacteria Crust
Cyanolichen Crust
Chlorolichen Crust
Moss Crust
Biofilm
RI_Mean
1.0
1.2
1.4
1.6
NRI_Mean
0.0
0.1
0.2
0.3
0.4
PNRI_Mean
0
10
20
30
40
Biocrust Coarse
Biofilm
Moss Crust
Chlorolichen Crust
Cyanolichen Crust
Dark Cyanobacteria Crust
Light Cyanobacteria Crust

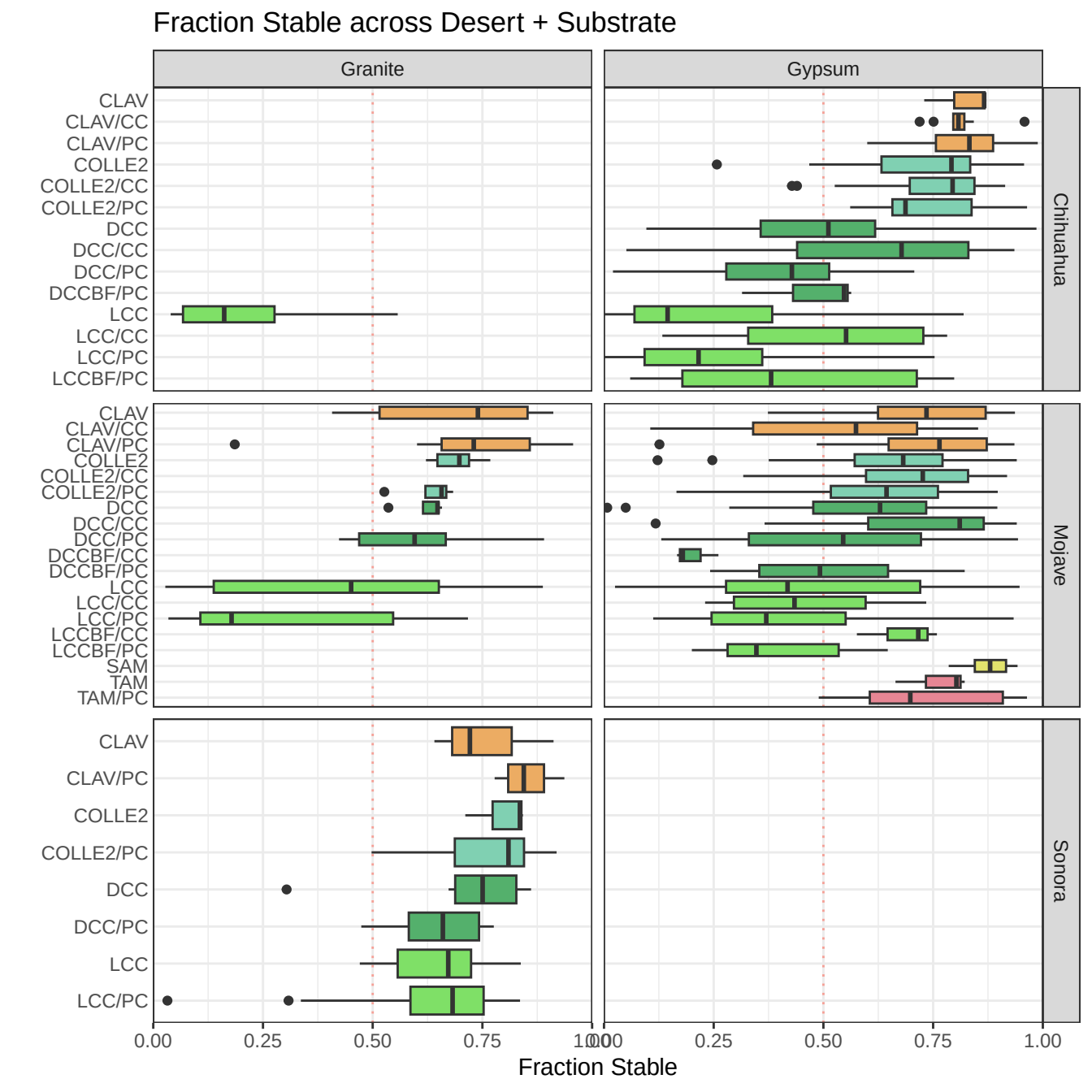
Roughness Indices Fine

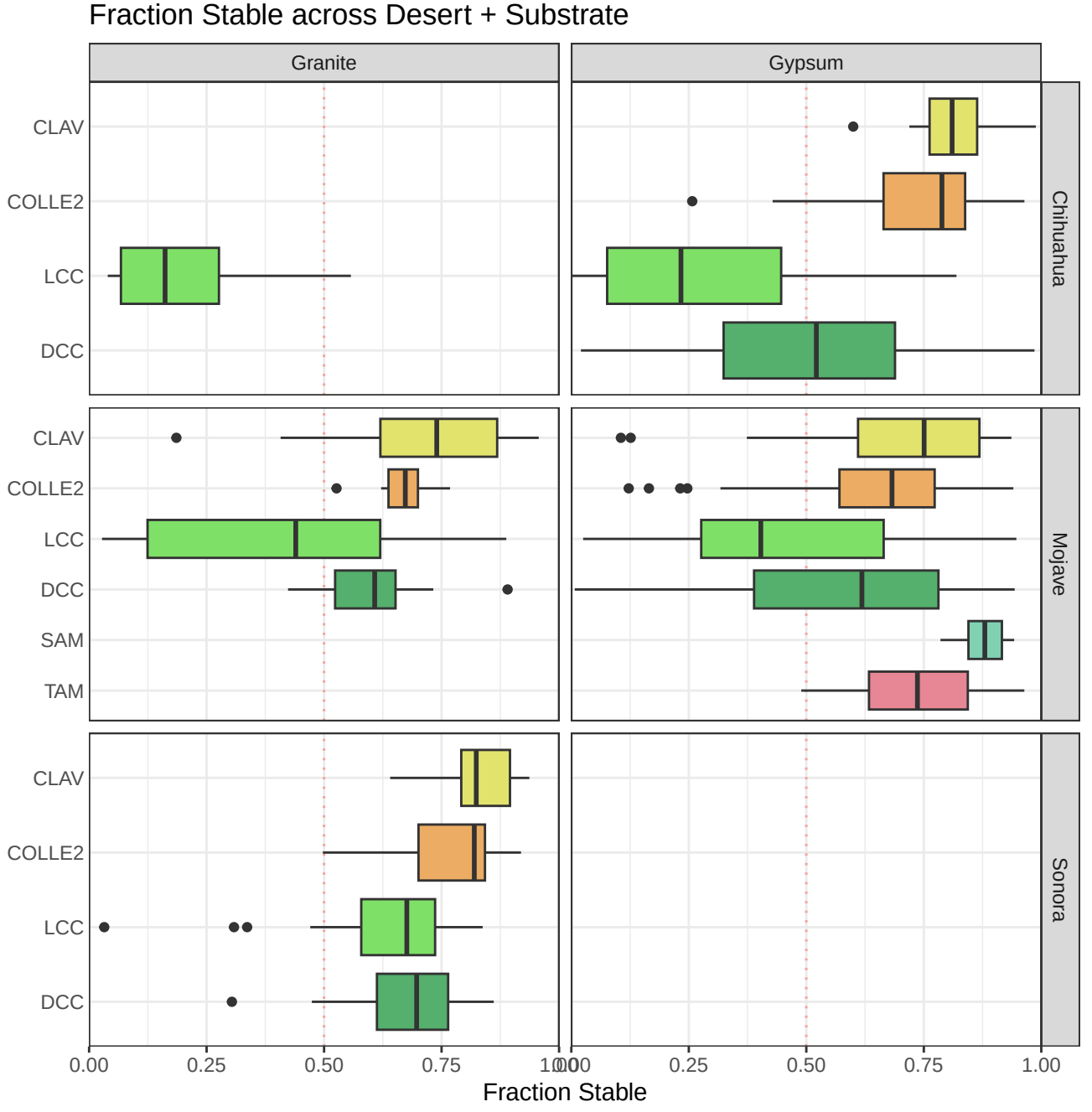


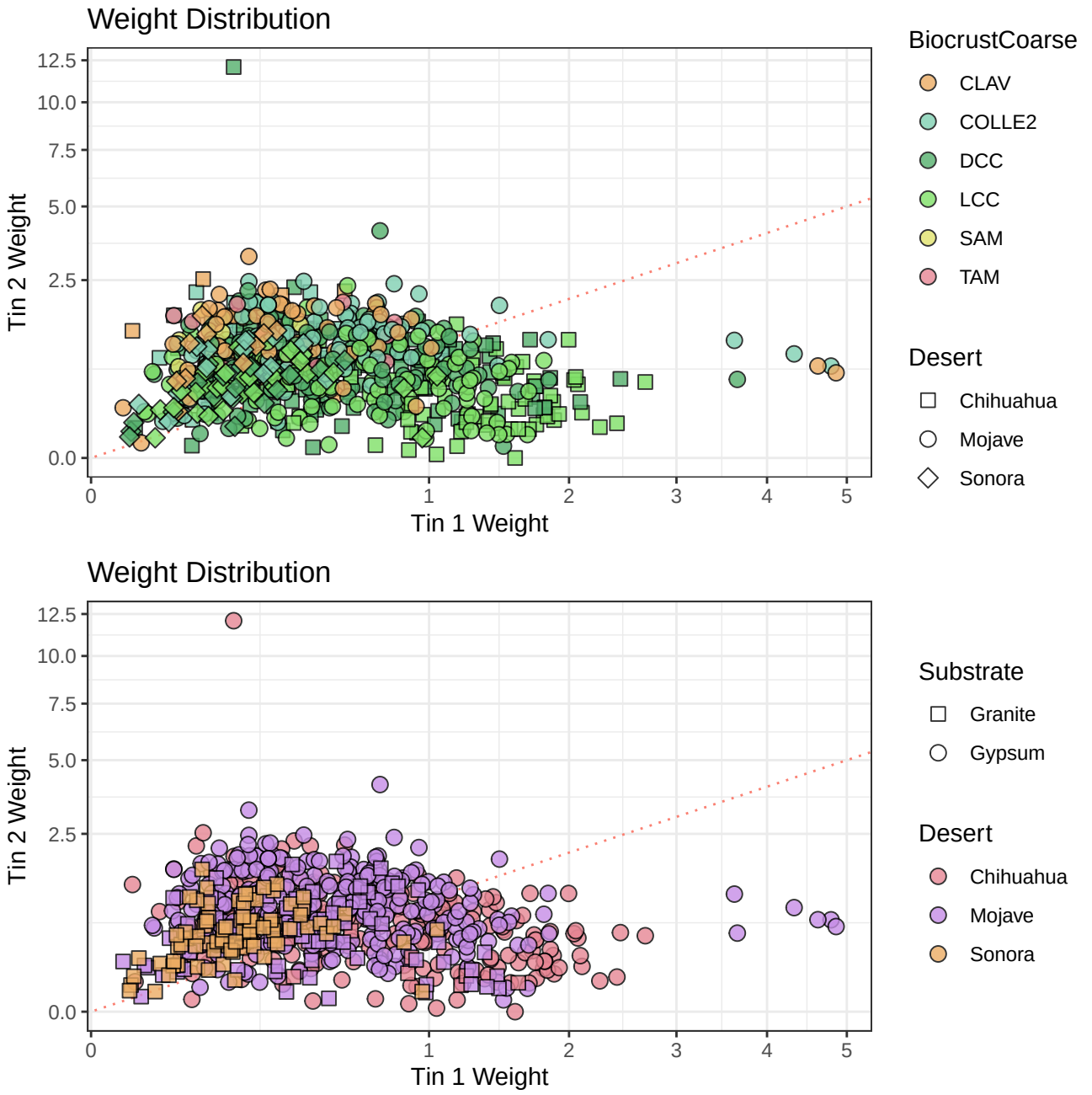

RI_Mean
NRI_Mean
PNRI_Mean
1.6
1.4
1.2
1.0
0.4
0.3
0.2
0.1
0.0
40
30
20
10
0
Chihuahua
Mojave
Sonora
Desert
Chihuahua
Mojave
Sonora

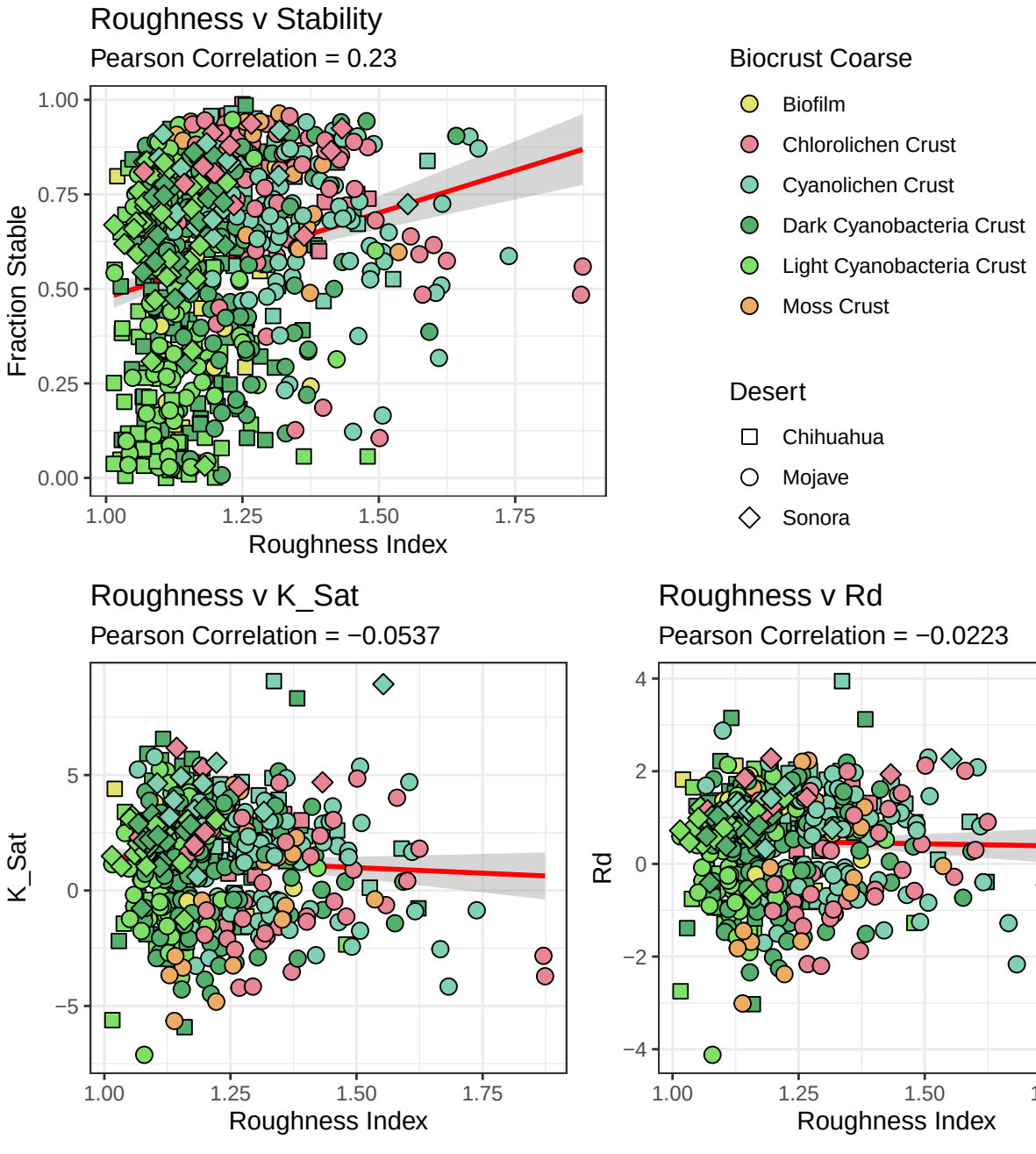

RI_Mean
NRI_Mean
PNRI_Mean
1.6
1.4
1.2
1.0
0.4
0.3
0.2
0.1
0.0
40
30
20
10
0
Chihuahua
Mojave
Sonora

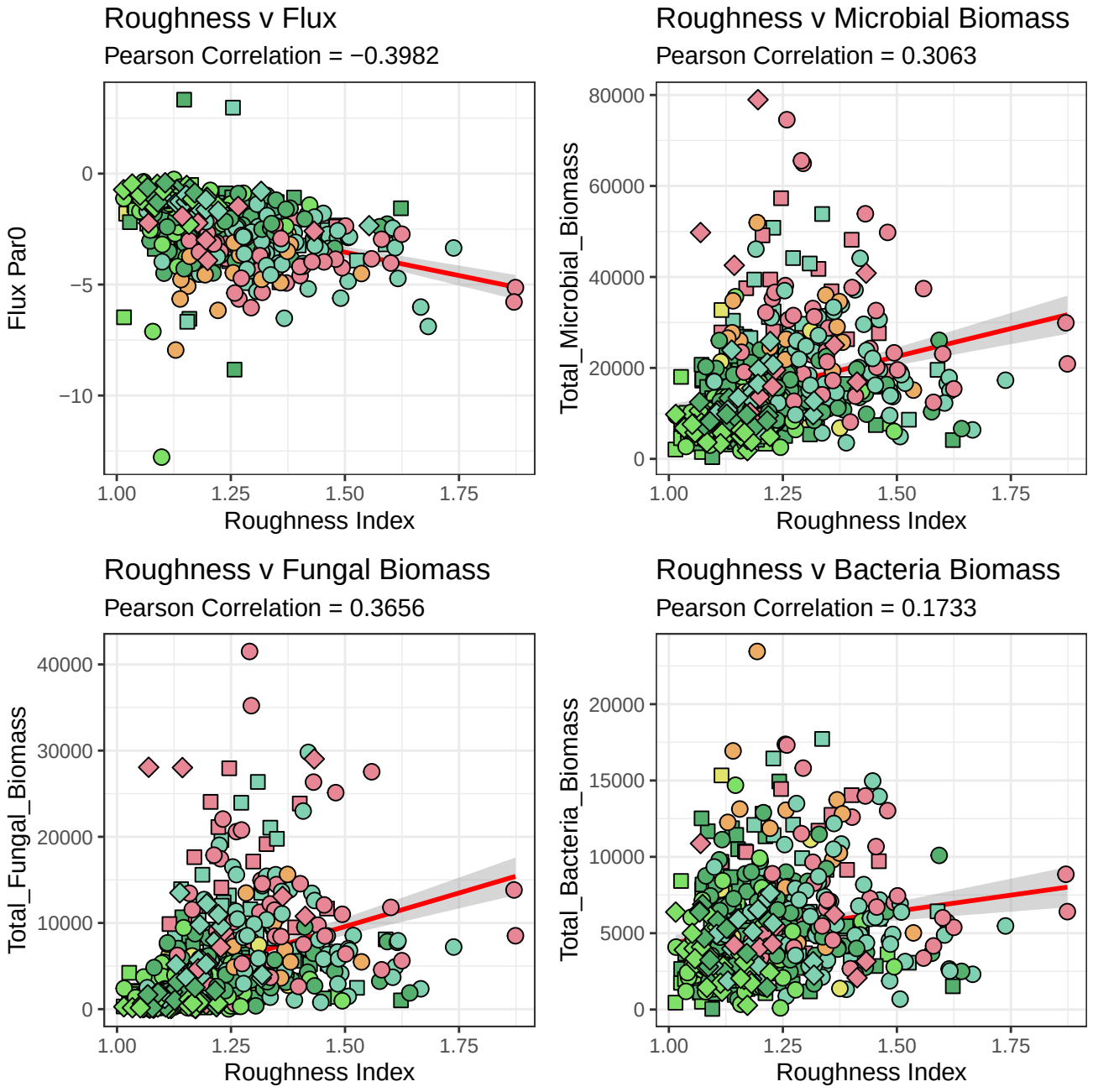

RI_Mean
NRI_Mean
PNRI_Mean
1.6
1.4
1.2
1.0
0.4
0.3
0.2
0.1
0.0
40
30
20
10
0
Chihuahua
Mojave
Sonora
Substrate
Granite
Gypsum


Fraction Stable across Desert + Substrate
Granite
Gypsum
Chihuahua
Mojave
Sonora
CLAV
CLAV/CC
CLAV/PC
COLLE2
COLLE2/CC
COLLE2/PC
DCC
DCC/CC
DCC/PC
DCCBF/CC
DCCBF/PC
LCC
LCC/CC
LCC/PC
LCCBF/CC
LCCBF/PC
SAM
TAM
TAM/PC
0.00
0.25
0.50
0.75
1.00
Fraction Stable


Fraction Stable across Desert + Substrate
Granite
Gypsum
Chihuahua
Mojave
Sonora
CLAV
COLLE2
LCC
DCC
SAM
TAM
0.00
0.25
0.50
0.75
1.00
Fraction Stable

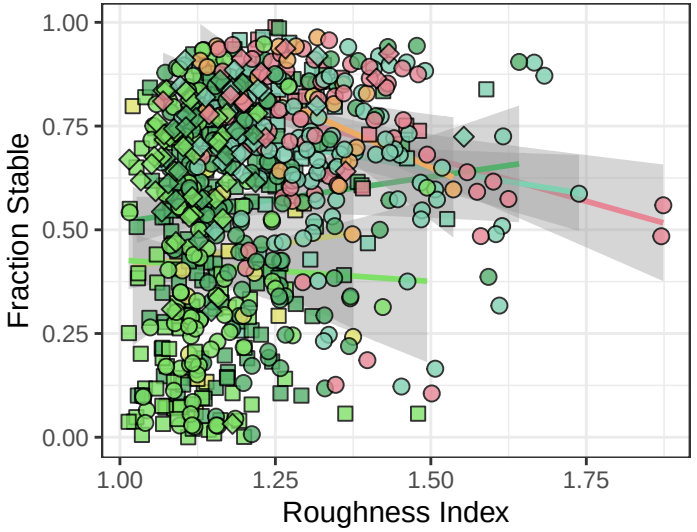

Weight Distribution
Tin 2 Weight
Tin 1 Weight
12.5
10.0
7.5
5.0
2.5
0.0
0
1
2
3
4
5
BiocrustCoarse
CLAV
COLLE2
DCC
LCC
SAM
TAM
Desert
Chihuahua
Mojave
Sonora
Weight Distribution
Tin 2 Weight
Tin 1 Weight
Substrate
Granite
Gypsum
Desert
Chihuahua
Mojave
Sonora


Roughness v Stability
Pearson Correlation = 0.23
Fraction Stable
Roughness Index
1.00
0.75
0.50
0.25
0.00
1.00
1.25
1.50
1.75
Biocrust Coarse
Biofilm
Chlorolichen Crust
Cyanolichen Crust
Dark Cyanobacteria Crust
Light Cyanobacteria Crust
Moss Crust
Desert
Chihuahua
Mojave
Sonora
Roughness v K_Sat
Pearson Correlation = −0.0537
K_Sat
Roughness Index
5
0
−5
1.00
1.25
1.50
1.75
Roughness v Rd
Pearson Correlation = −0.0223
Rd
Roughness Index
4
2
0
−2
−4
1.00
1.25
1.50
1.75


Roughness v Flux
Pearson Correlation = −0.3982
Flux Par0
0
−5
−10
1.00
1.25
1.50
1.75
Roughness Index
Roughness v Microbial Biomass
Pearson Correlation = 0.3063
Total_Microbial_Biomass
80000
60000
40000
20000
0
1.00
1.25
1.50
1.75
Roughness Index
Roughness v Fungal Biomass
Pearson Correlation = 0.3656
Total_Fungal_Biomass
40000
30000
20000
10000
0
1.00
1.25
1.50
1.75
Roughness Index
Roughness v Bacteria Biomass
Pearson Correlation = 0.1733
Total_Bacteria_Biomass
20000
15000
10000
5000
0
1.00
1.25
1.50
1.75
Roughness Index

Roughness v Stability

Pearson Correlations:
Biofilm=0.091; Chlorolichen Crust=−0.341; Cyanolichen Crust=−0.122
Dark Cyanobacteria Crust=0.149; Light Cyanobacteria Crust=0.004; Moss Crust=−0.533



Biocrust Coarse

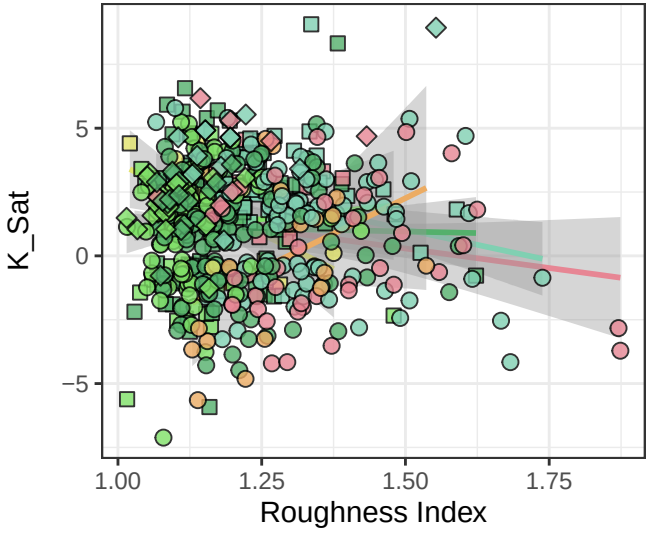
- Biofilm
- Chlorolichen Crust
- Cyanolichen Crust
- Dark Cyanobacteria Crust
- Light Cyanobacteria Crust
- Moss Crust

Desert

- Chihuahua
- Mojave
- Sonora

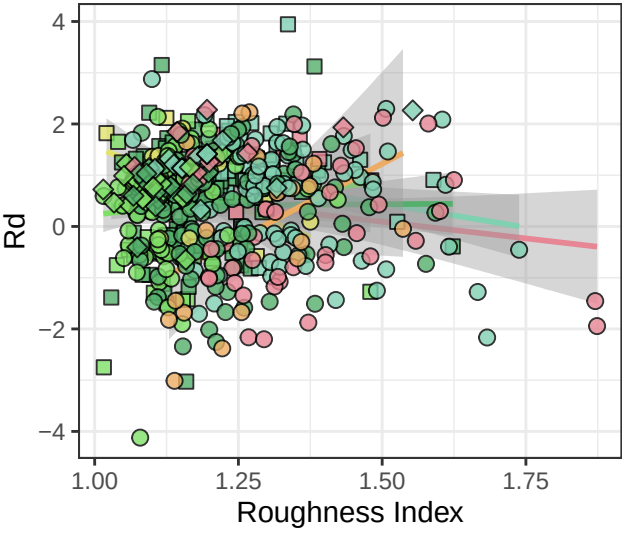
Roughness v K_Sat

Pearson Correlations:
Biofilm=−0.514; Chlorolichen Crust=−0.189; Cyanolichen Crust=−0.261
Dark Cyanobacteria Crust=−0.021; Light Cyanobacteria Crust=0.076; Moss Crust=0.388



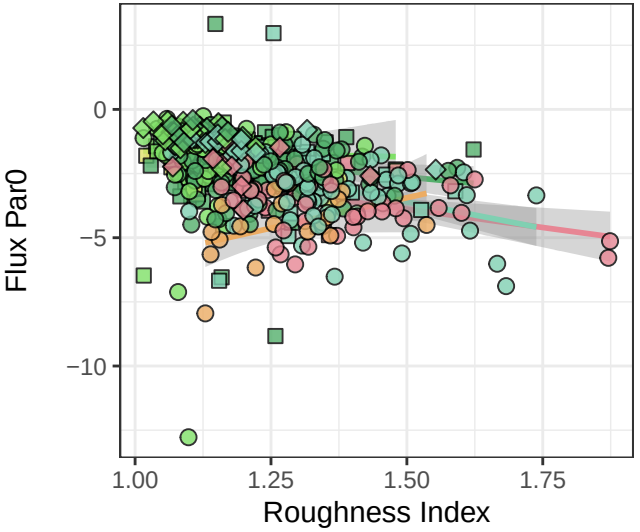
Roughness v Rd

Pearson Correlations:
Biofilm=−0.521; Chlorolichen Crust=−0.172; Cyanolichen Crust=−0.237
Dark Cyanobacteria Crust=0.007; Light Cyanobacteria Crust=0.092; Moss Crust=0.402



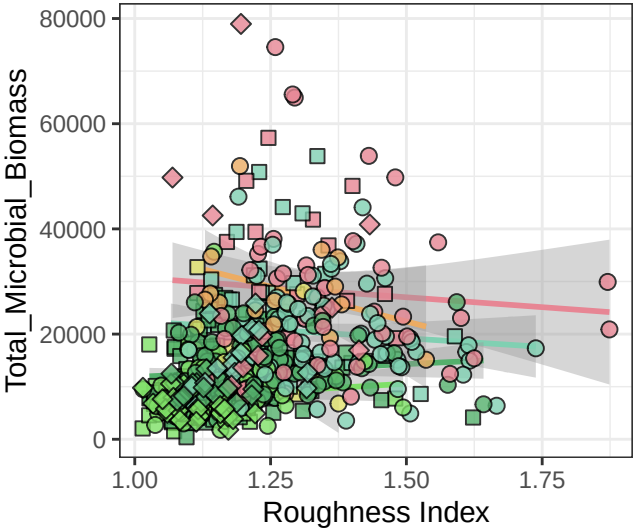
Roughness v Flux

Pearson Correlations:
Biofilm=−0.403; Chlorolichen Crust=−0.403; Cyanolichen Crust=−0.411
Dark Cyanobacteria Crust=−0.182; Light Cyanobacteria Crust=−0.024; Moss Crust=0.426



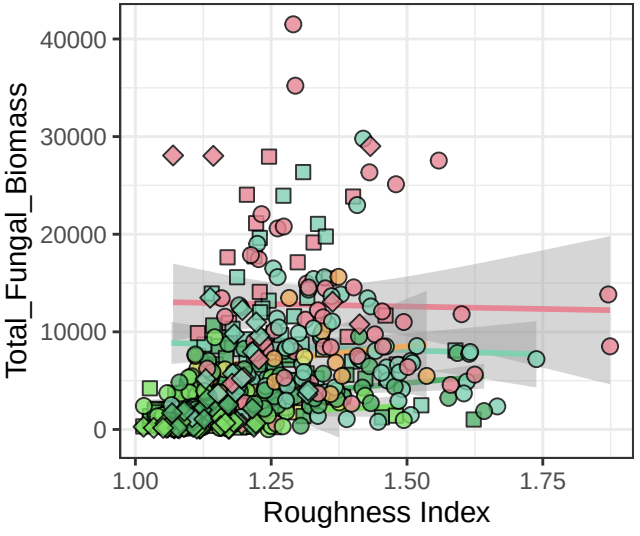
Roughness v Microbial Biomass

Pearson Correlations:
Biofilm=−0.234; Chlorolichen Crust=−0.073; Cyanolichen Crust=−0.089
Dark Cyanobacteria Crust=0.092; Light Cyanobacteria Crust=0.109; Moss Crust=−0.361



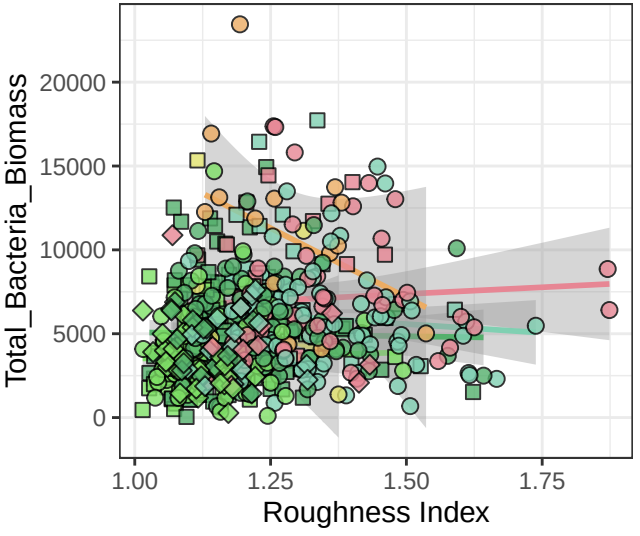
Roughness v Fungal Biomass

Pearson Correlations:
Biofilm=−0.237; Chlorolichen Crust=−0.018; Cyanolichen Crust=−0.041
Dark Cyanobacteria Crust=0.239; Light Cyanobacteria Crust=0.134; Moss Crust=0.176



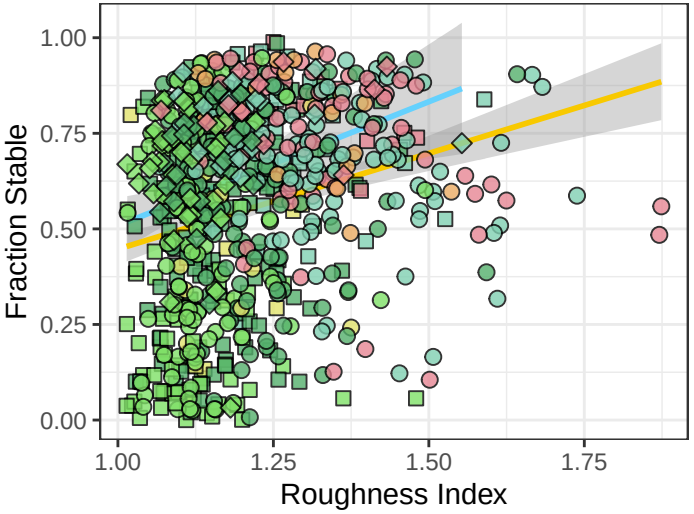
Roughness v Bacteria Biomass

Pearson Correlations:
Biofilm=−0.238; Chlorolichen Crust=0.063; Cyanolichen Crust=−0.087
Dark Cyanobacteria Crust=−0.019; Light Cyanobacteria Crust=0.027; Moss Crust=−0.364



Roughness v Stability

Pearson Correlations:
Granite=0.236; Gypsum=0.262



Substrate

Granite

Gypsum

Biocrust Coarse

Biofilm

Chlorolichen Crust

Cyanolichen Crust

Dark Cyanobacteria Crust

Light Cyanobacteria Crust

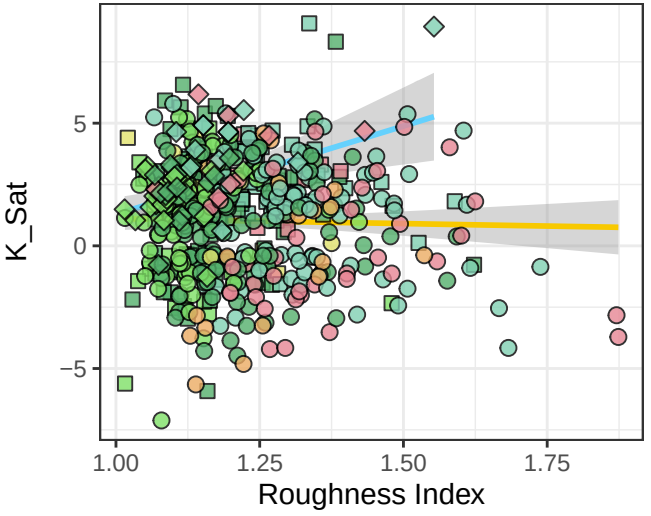
Moss Crust

Desert

Chihuahua

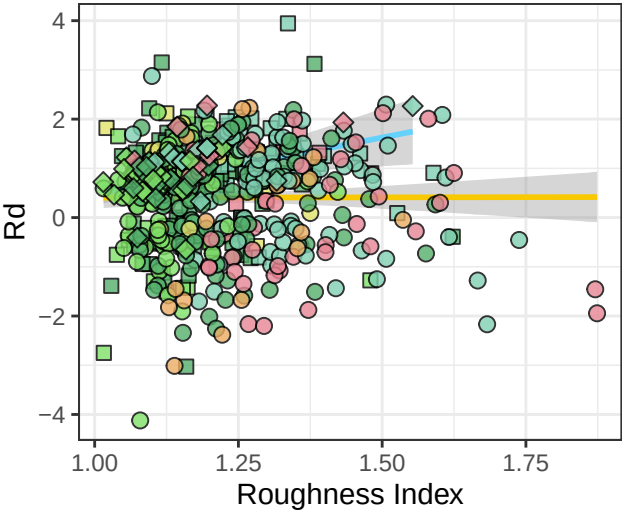
Roughness v K_Sat

Pearson Correlations:
Granite=0.317; Gypsum=−0.022



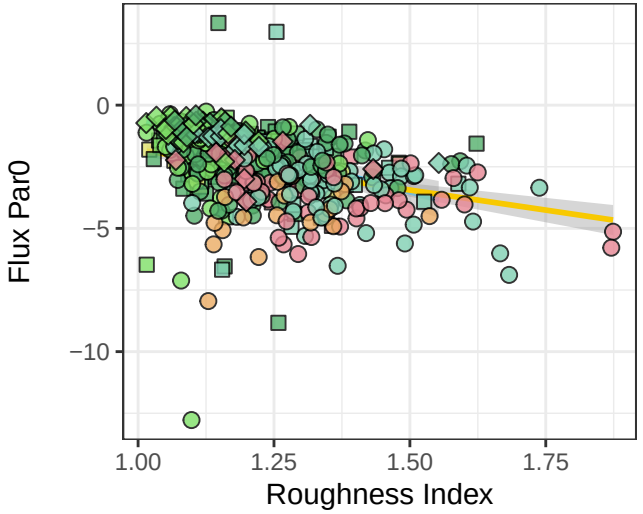
Roughness v Rd

Pearson Correlations:
Granite=0.275; Gypsum=0.002



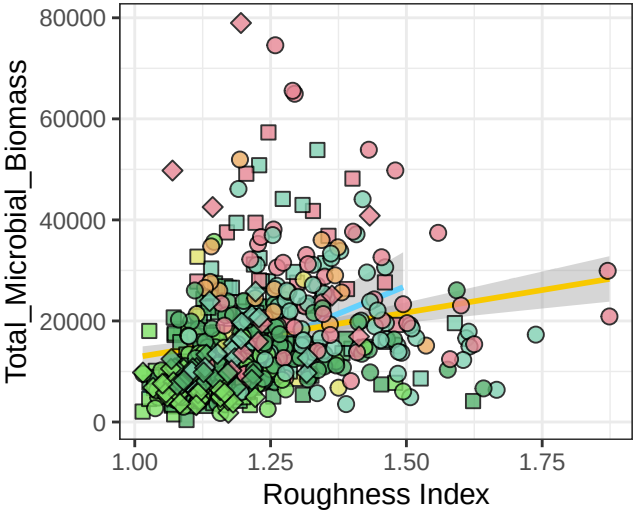
Roughness v Flux

Pearson Correlations:
Granite=−0.542; Gypsum=−0.312



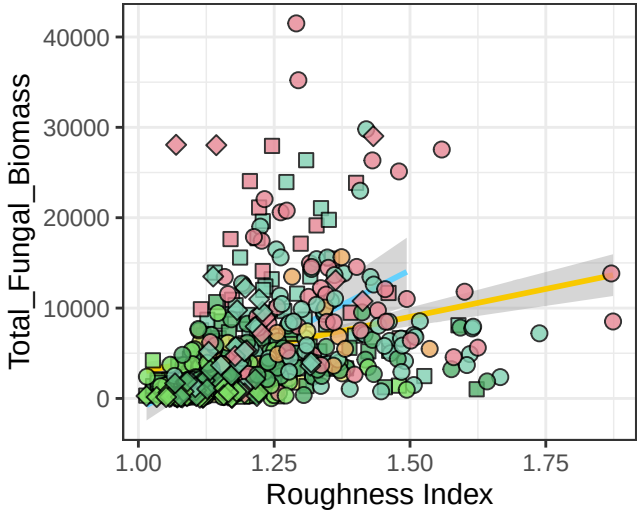
Roughness v Microbial Biomass

Pearson Correlations:
Granite=0.351; Gypsum=0.237



Roughness v Fungal Biomass

Pearson Correlations:
Granite=0.433; Gypsum=0.313



Roughness v Bacteria Biomass

Pearson Correlations:
Granite=0.066; Gypsum=0.113

